Project Report: Predicting and Explaining Credit Risk with Machine Learning

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1. Executive Summary

The Business Problem: The core challenge in banking is the "Credit Paradox": lending too cautiously results in lost business (opportunity cost), while lending too aggressively leads to massive financial losses from defaults. The goal is to find the optimal balance approving as many creditworthy applicants as possible while minimizing losses from those who will default.

Our Proposed Solution: We developed an end-to-end machine learning framework to address this challenge. This project involved building a predictive model that calculates a risk score (probability of default) for each loan applicant. Crucially, the framework goes beyond simple prediction by providing clear, human-understandable explanations for each decision, ensuring transparency and regulatory compliance.

Key Achievements & Results:

- **High Predictive Power:** The final LightGBM model achieved an **Area Under the Curve (AUC) of 0.77** on unseen data, demonstrating a strong ability to distinguish between safe and risky applicants.
- **Effective Risk Identification:** By strategically weighting the minority class (defaulters), the model was tuned to successfully identify **69% of actual defaulters (Recall)**, directly addressing the primary business goal of reducing credit losses.
- **Actionable Insights:** Model analysis revealed that the most critical predictors of default are **external credit bureau scores (EXT_SOURCE_2 & 3)**, the applicant's age (YEARS_BIRTH), and engineered financial strain metrics like the **Annuity-to-Credit Ratio**.
- **Full Explainability (XAI):** Using the SHAP framework, we can now explain the exact drivers behind any individual's risk score, moving from a "black box" prediction to a transparent, auditable decision-making tool.

Business Impact: This project provides a blueprint for a data-driven lending system that can reduce credit losses, increase the efficiency of loan officers, and ensure fair and consistent lending decisions. It transforms the risk assessment process from a subjective evaluation into a precise, data-backed science.

2. The Business Problem & Project Objective

The project tackles the fundamental business problem of **Credit Risk Assessment**. Home Credit, like any financial institution, needs to accurately evaluate the risk associated with each loan application. A wrong decision has two potential negative outcomes:

1. **Approving a High-Risk Applicant:** This leads to a loan default, resulting in a direct financial loss for the company.

2. **Rejecting a Low-Risk Applicant:** This results in lost revenue and potential loss of the customer to a competitor, known as an opportunity cost.

The primary objective was to build a machine learning model that could solve a **binary classification problem**:

- **Input:** Applicant's financial and demographic information.
- **Output:** A probability score indicating the likelihood of the applicant defaulting on the loan.
- **Success Metric (Technical):** Maximize the **AUC score** to ensure the model has strong discriminatory power.
- **Success Metric (Business):** Maximize **Recall** for the 'Default' class to catch as many potential defaulters as possible, thereby minimizing credit loss.

3. Tools & Technologies Used

- **Language:** Python 3
- **Core Libraries:** Pandas (Data Manipulation), NumPy (Numerical Operations), Scikit-learn (Preprocessing & Metrics)
- **Machine Learning:** LightGBM (Gradient Boosting Model)
- **Data Visualization:** Matplotlib, Seaborn
- **Model Explainability:** SHAP (SHapley Additive exPlanations)
- **Environment:** Kaggle Notebooks

4. Methodology and Technical Implementation

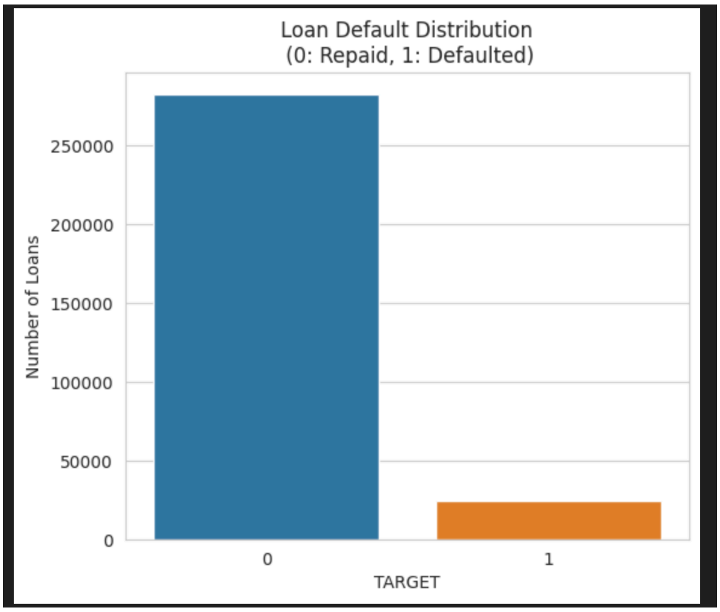
This project followed a structured, end-to-end data science lifecycle.

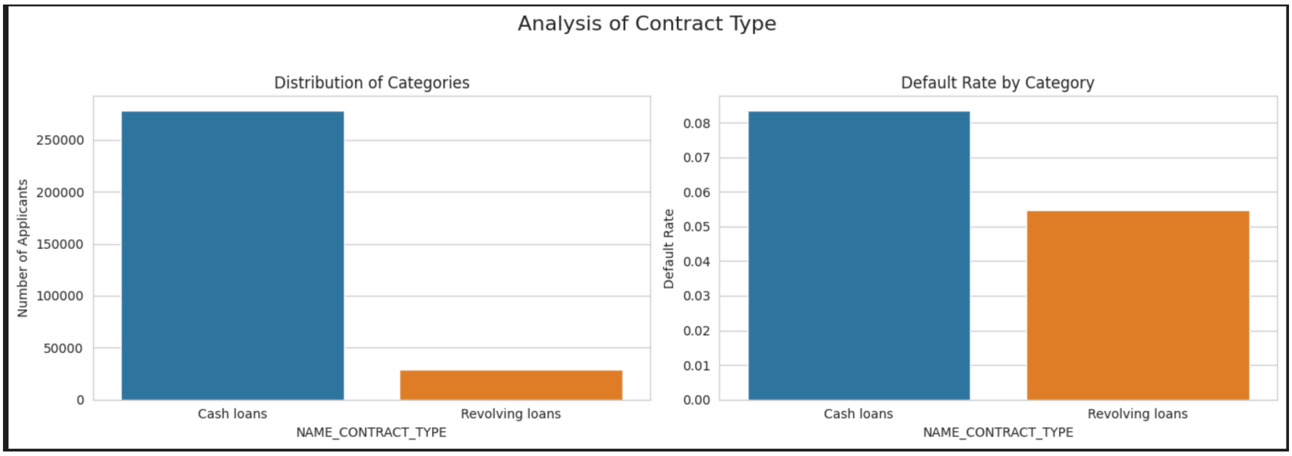
4.1. Exploratory Data Analysis (EDA)

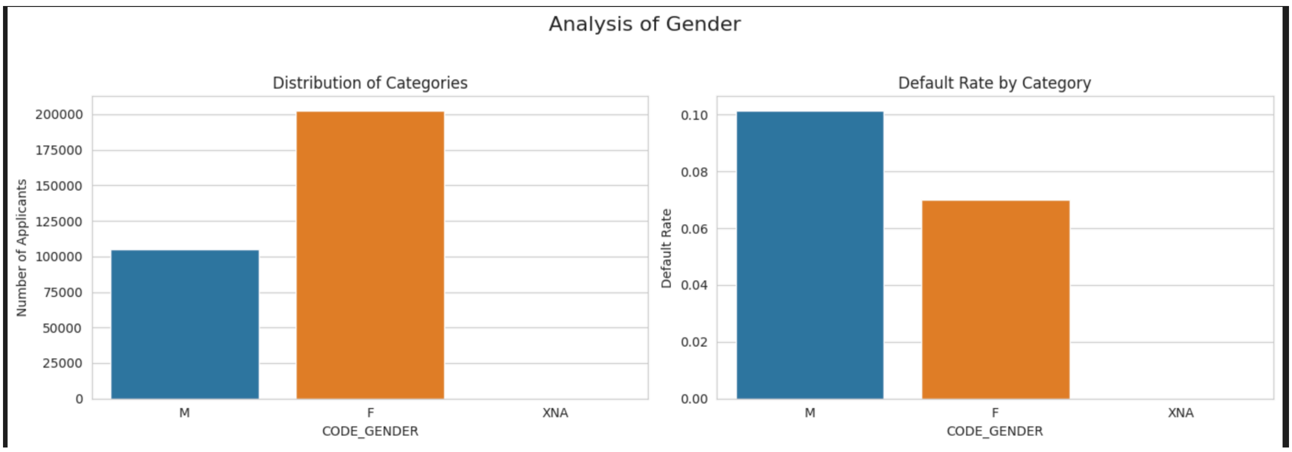
Initial analysis of the 307,511 loan applications revealed several critical insights that guided the entire project:

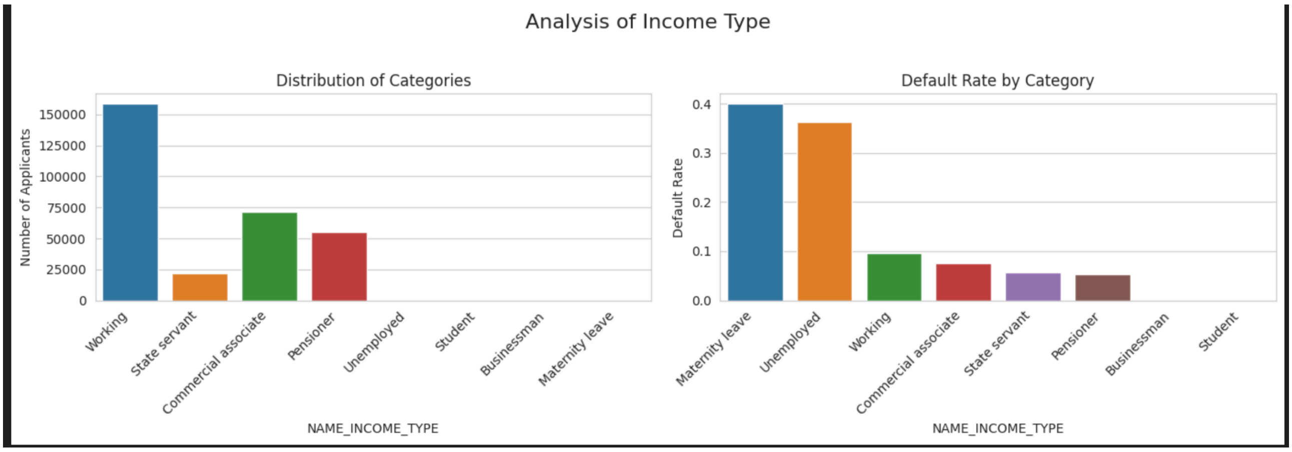
- **Severe Class Imbalance:** The dataset was highly imbalanced, with only **8% of applicants defaulting**. This meant that 'accuracy' would be a misleading metric and specialized techniques would be required.
- **Risk Profiling:** Analysis showed distinct patterns. Higher default rates were observed among applicants who were younger, had less work experience, were single, and were male.
- **Data Anomaly Detection:** A significant anomaly was discovered in the DAYS_EMPLOYED feature, where a value of 365243 appeared for over 55,000 applicants. Further investigation revealed these applicants had a *lower* default rate, suggesting they

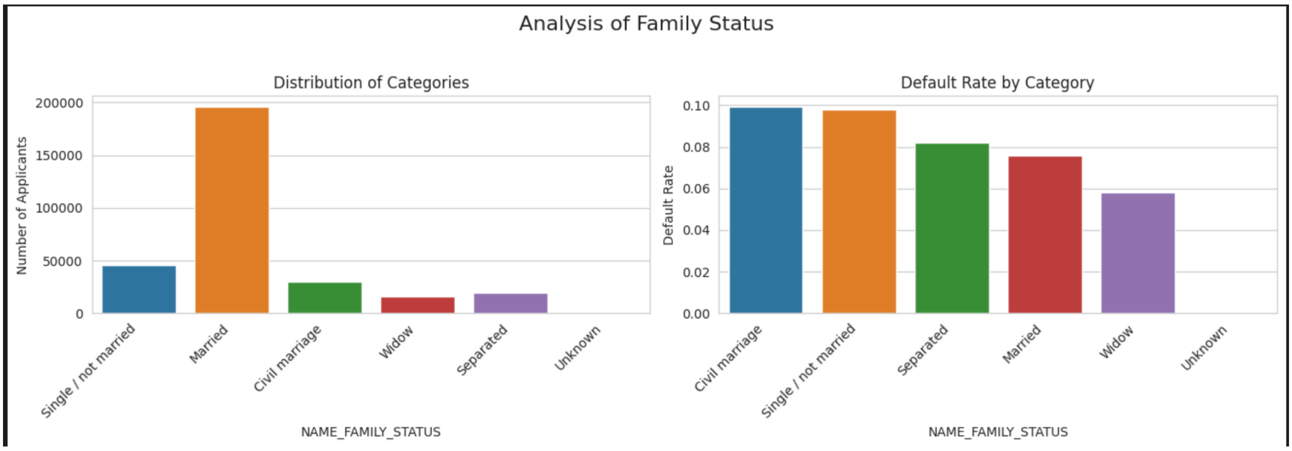
represented a distinct, stable group (e.g., retirees). This anomaly was converted into a valuable predictive feature.

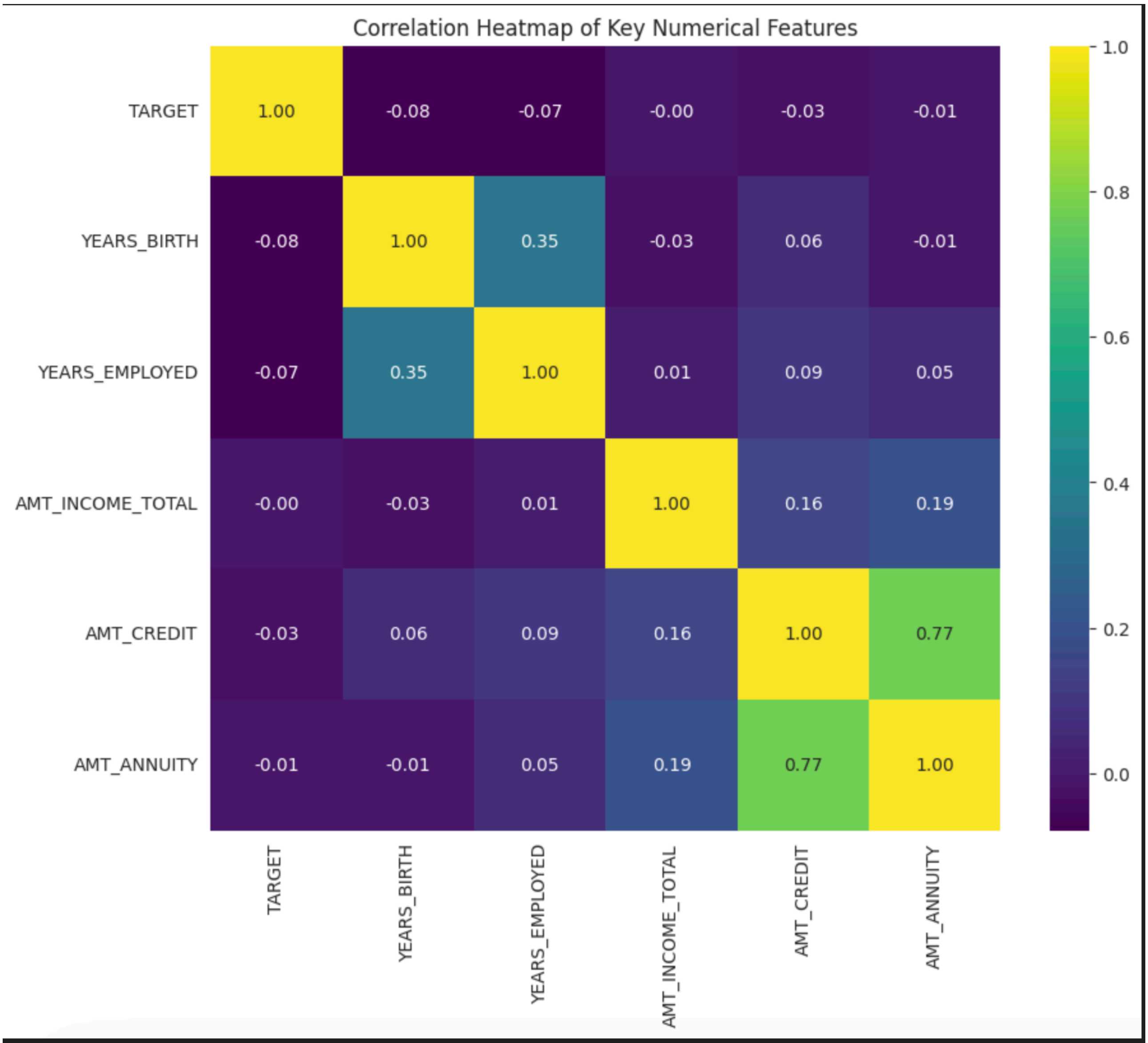







Analysis of Income Type
Distribution of Categories
Number of Applicants
150000
125000
100000
75000
50000
25000
0
Working
State servant
Commercial associate
Pensioner
Unemployed
Student
Businessman
Maternity leave
NAME_INCOME_TYPE
Default Rate by Category
Default Rate
0.4
0.3
0.2
0.1
0.0
Maternity leave
Unemployed
Working
Commercial associate
State servant
Pensioner
Businessman
Student
NAME_INCOME_TYPE


Analysis of Family Status
Distribution of Categories
Number of Applicants
200000
150000
100000
50000
0
Single / not married
Married
Civil marriage
Widow
Separated
Unknown
NAME_FAMILY_STATUS
Default Rate by Category
Default Rate
0.10
0.08
0.06
0.04
0.02
0.00
Civil marriage
Single / not married
Separated
Married
Widow
Unknown
NAME_FAMILY_STATUS


Correlation Heatmap of Key Numerical Features
TARGET 1.00 -0.08 -0.07 -0.00 -0.03 -0.01
YEARS_BIRTH -0.08 1.00 0.35 -0.03 0.06 -0.01
YEARS_EMPLOYED -0.07 0.35 1.00 0.01 0.09 0.05
AMT_INCOME_TOTAL -0.00 -0.03 0.01 1.00 0.16 0.19
AMT_CREDIT -0.03 0.06 0.09 0.16 1.00 0.77
AMT_ANNUITY -0.01 -0.01 0.05 0.19 0.77 1.00
TARGET
YEARS_BIRTH
YEARS_EMPLOYED
AMT_INCOME_TOTAL
AMT_CREDIT
AMT_ANNUITY
1.0
0.8
0.6
0.4
0.2
0.0

4.2. Feature Engineering & Data Preprocessing

Raw data is rarely optimal for a machine learning model. The following steps were taken to create a rich, clean feature set:

- **Domain-Specific Feature Creation:** Based on business logic, new, powerful features were engineered to capture an applicant's financial health:
  - CREDIT_INCOME_PERCENT: Measures the loan amount relative to income.
  - ANNUITY_INCOME_PERCENT: Measures the burden of monthly payments on income.
  - ANNUITY_CREDIT_RATIO: Approximates the loan's repayment term/speed. These engineered features proved to be among the most important in the final model.
- **Handling Missing Values:** A strategic approach was used:
  - **Categorical Features:** NaNs were filled with a constant 'Unknown' category, allowing the model to learn from the pattern of missingness itself.
  - **Numerical Features:** NaNs were imputed with the **median** value from the training set. The median was chosen over the mean due to its robustness to outliers common in financial data.
- **Categorical Encoding:** All object-type columns were transformed into a numerical format using **One-Hot Encoding** to prevent the model from inferring an artificial ordinal relationship between categories.

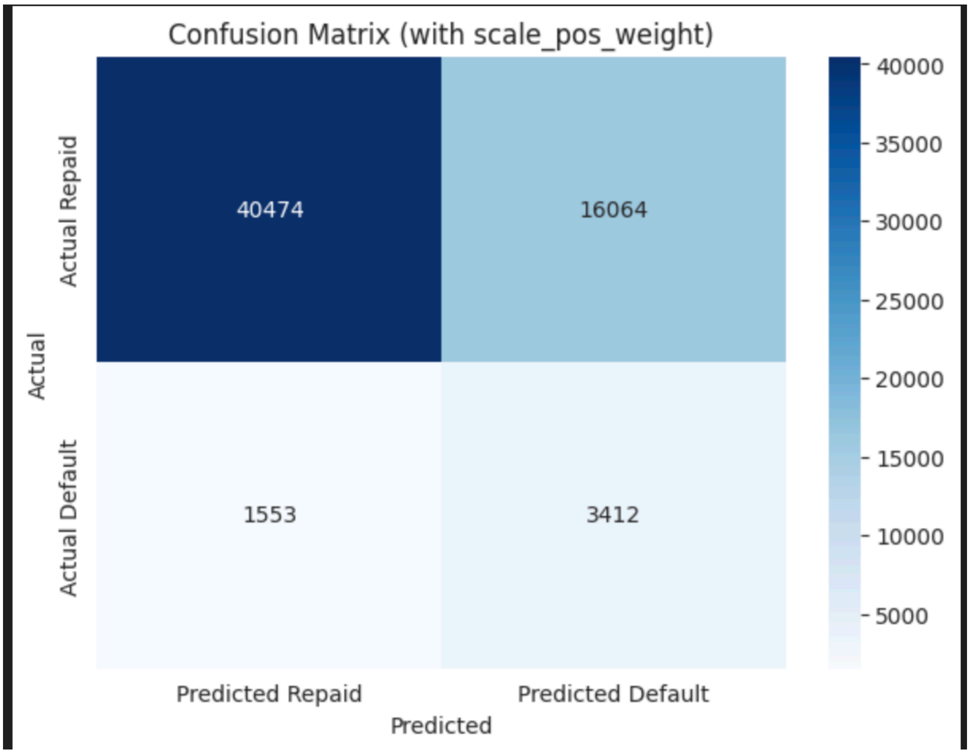
4.3. Modeling & Evaluation

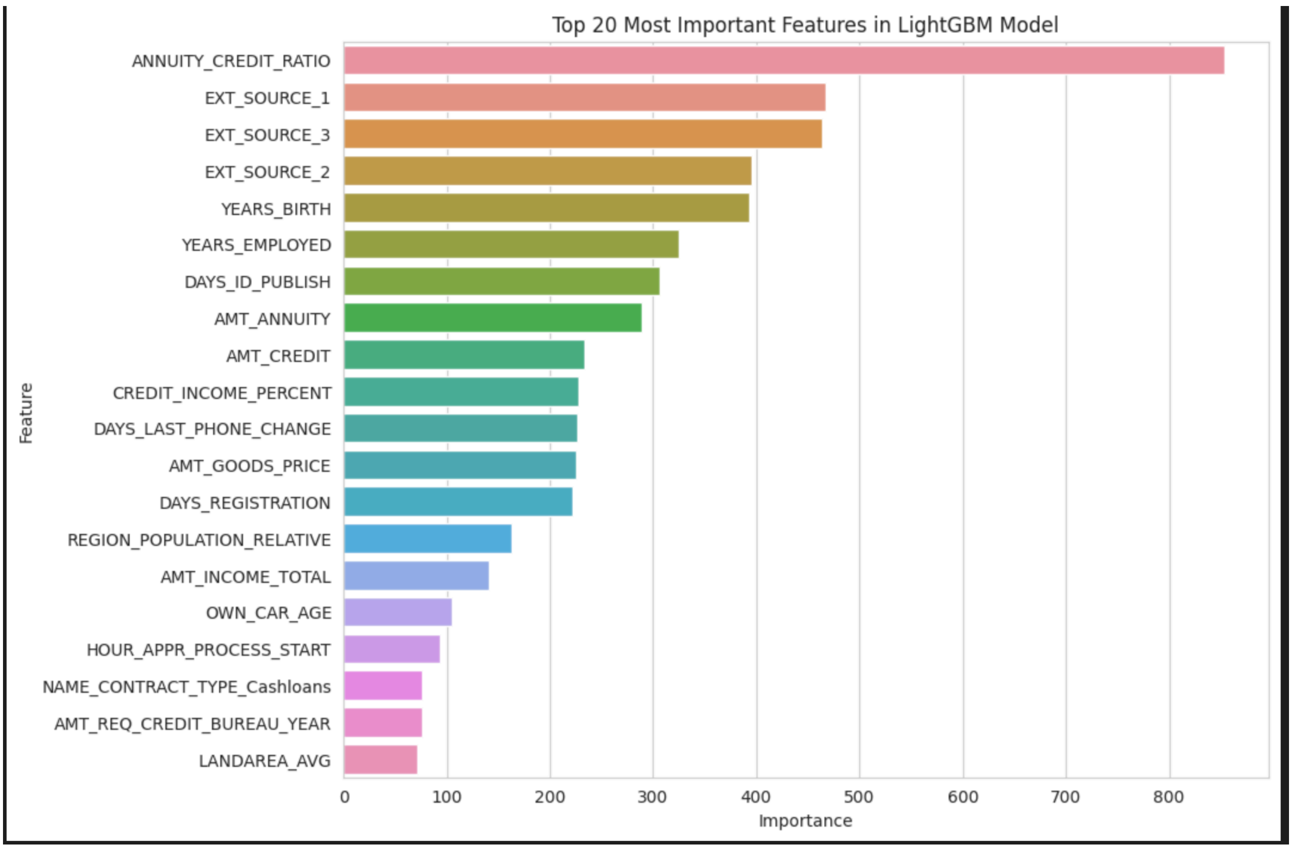
- **Model Selection: LightGBM**, a gradient boosting framework, was selected for its high performance, speed on large datasets, and efficiency.
- **Addressing Class Imbalance:** Instead of simpler methods, the scale_pos_weight parameter was implemented. This was calculated as the ratio of non-defaulters to defaulters (~11.5). This technique forces the model to heavily penalize mistakes made on the minority (defaulter) class, thereby directly optimizing for higher recall.
- **Training & Validation:** The data was split into an 80% training set and a 20% validation set. The model was trained with **early stopping** for 100 rounds, which prevents overfitting by stopping the training process when the validation AUC score no longer improves.

Model Results:
The model achieved a final **Validation AUC of 0.7697**. The confusion matrix and classification report confirmed our strategy was successful:

- **Recall (Defaulted): 0.69** - The model successfully identified 69% of all clients who would eventually default.

- **Precision (Defaulted): 0.18** - When the model predicts a default, it is correct 18% of the time. While low, this is a significant lift over the 8% base rate and reflects the business trade-off to prioritize catching defaulters.
- **False Negatives (The Costliest Error):** This number was significantly reduced due to our high-recall strategy, directly translating to minimized credit loss.

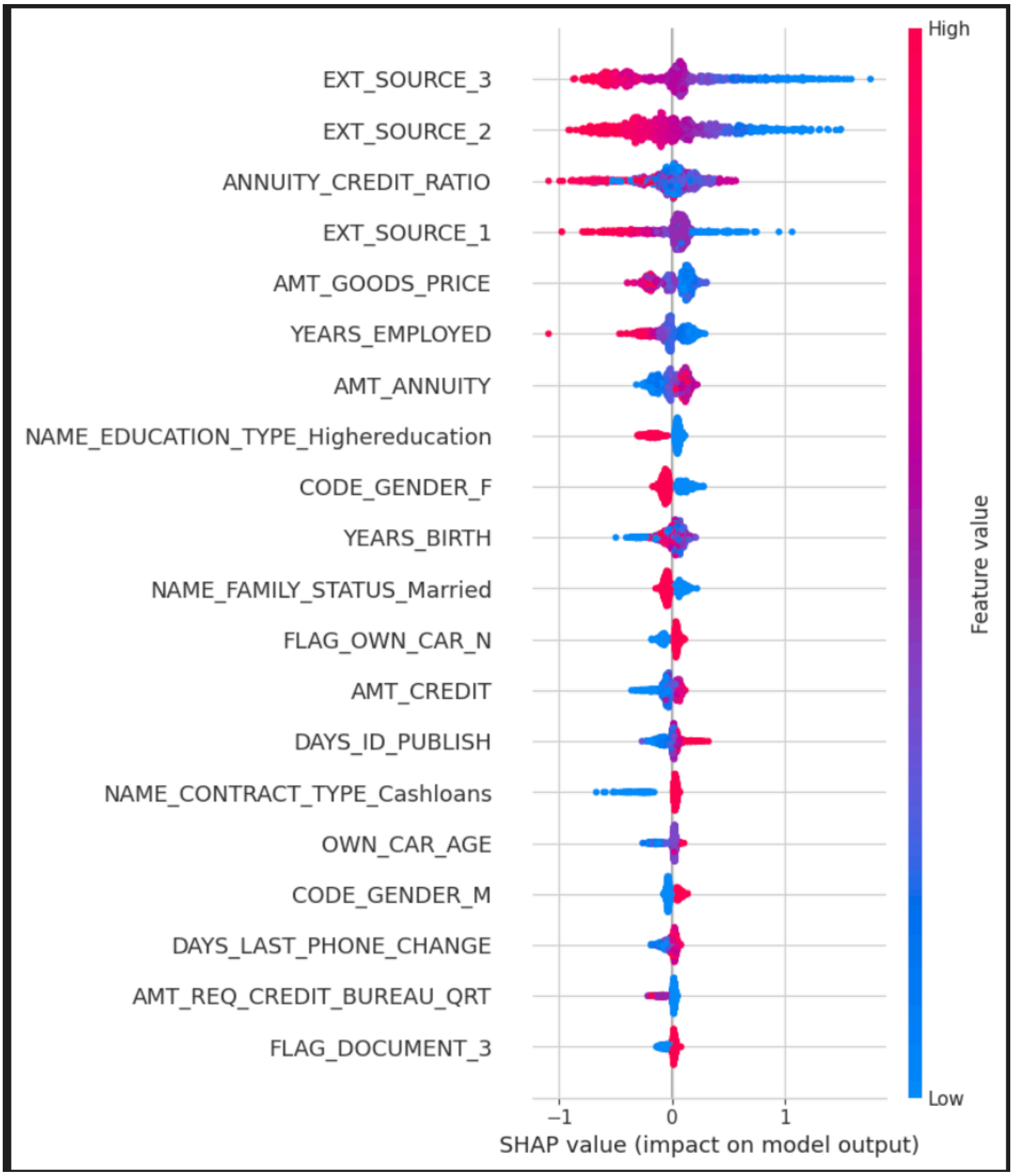




5. Model Explainability with SHAP (XAI)

A prediction is useless without an explanation. We used the SHAP library to understand the model's logic.

- **Global Importance:** The SHAP summary plot confirmed our EDA and business intuition. The most impactful features globally were external credit scores (EXT_SOURCE_2 & 3), YEARS_EMOLOYERS, ANNUITY_CREDIT_RATIO, and other key financial indicators.
- **Local Explainability:** We generated force plots for individual predictions. For example, we analyzed a "False Negative" (a high-risk applicant the model missed). The SHAP plot revealed that while many features pointed towards low risk (e.g., stable income), a few highly negative features (like a very poor external credit score) were the true drivers of risk, which the model didn't weigh heavily enough. This level of granular analysis is invaluable for model debugging and building trust with loan officers.



6. Challenges Faced & Solutions

1. **Challenge:** Severe class imbalance risked creating a useless model that always predicted "repay".
    - **Solution:** Implemented the scale_pos_weight parameter in LightGBM, making a conscious business decision to trade lower precision for a much higher, more valuable recall rate.
2. **Challenge:** A cryptic anomaly in the DAYS_EMPLOYED column could have been discarded.
    - **Solution:** Investigated the anomaly, hypothesized its meaning (retirees), and engineered a new binary feature that improved model performance.
3. **Challenge:** The LightGBM library threw a LightGBMError due to special JSON characters in column names created during one-hot encoding.
    - **Solution:** Wrote a Python function using regular expressions (re.sub) to programmatically sanitize all column names, making the data pipeline robust and compatible with the modeling library.

7. Conclusion & Future Work

This project successfully demonstrates the development of a robust and explainable machine learning model for credit risk assessment. The model not only delivers strong predictive performance but also provides the transparency required in a modern, regulated financial environment.

Recommendations & Next Steps:

- **Integrate Additional Data:** The next major performance gain will come from joining the auxiliary datasets provided (bureau.csv, previous_application.csv), which contain rich information about applicants' credit history.
- **Hyperparameter Tuning:** Employ automated tuning libraries like Optuna or GridSearch to systematically find the optimal set of model parameters and further boost AUC.
- **Deployment Strategy:** The model can be containerized (e.g., using Docker) and deployed as a REST API. This would allow internal bank applications to send applicant data and receive a risk score in real-time.
- **Monitoring:** Once deployed, the model's performance must be monitored for "concept drift" to ensure its predictions remain accurate as economic conditions and customer behaviors change over time.